George Koufie

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PROFESSIONAL SUMMARY

Infrastructure and cloud engineer with 6+ years of enterprise IT operations experience and hands-on expertise in Python automation, AWS, Terraform, Linux, monitoring, and reliability engineering. Experienced in troubleshooting infrastructure incidents, automating operational workflows, implementing observability and alerting, and validating disaster recovery through failure simulations. Built cloud reliability projects using Kubernetes, Prometheus, Grafana, and CI/CD pipelines, with measurable recovery and remediation outcomes.

TECHNICAL SKILLS

Programming & automation: Python, Bash, PowerShell, REST APIs, Terraform, SaltStack

Monitoring & reliability: Prometheus, Grafana, CloudWatch, SLI/SLOs, incident response, root-cause analysis

Reliability & automation**:** Prometheus, Grafana, SLI/SLOs, burn-rate alerting, incident response, root-cause analysis, RTO/RPO, failover, backup/recovery, Python, Bash, PowerShell, SaltStack, Linux, Windows Server

Cloud architecture: AWS: VPC, EC2, IAM, S3, RDS, EKS, Lambda, API Gateway, DynamoDB, CloudFront, CloudWatch, CloudTrail, Systems Manager; Azure: VMs, VNets, Entra ID, Azure SQL, Azure Site Recovery, Azure Monitor

AI & System Integration: Model Context Protocol (MCP), MCP Servers, AI Agents, LLM Tool Integration, REST APIs, API Integration, SQL, SQLite, Azure SQL, Cosmos DB

Infrastructure & delivery**:** Terraform, CloudFormation, Azure Bicep, GitHub Actions, Jenkins, Azure DevOps, CI/CD, GitOps, Helm, Argo CD, Docker, Kubernetes, EKS

Security & networking**:** IAM/RBAC, least privilege, OIDC, MFA, IMDSv2, patch baselines, SSM, OpenSCAP/STIG validation, VPN/IPsec, Transit Gateway, routing, DNS, TCP/IP, security groups, TLS

SELECTED PROJECTS

AWS SaaS SRE & Disaster Recovery Lab (EKS, RDS, S3, ECR, IAM), Kubernetes, Terraform, Helm, Prometheus/Grafana, Python, GitHub Actions Oct 2026

- Deployed a FastAPI/PostgreSQL application and simulated deletion of 166,000+ rows; completed RDS point-in-time restore in under 16 minutes.
- Implemented SLO burn-rate alerting, exercised five failure scenarios, documented blameless incident reviews, and remediated ten reliability issues; integrated Checkov/Trivy into CI. GitHub Repo

PatchPilot: Batch-Aware Patch Automation & Validation | Python, SaltStack, Terraform, AWS (EC2, Systems Manager, CloudWatch, S3), RHEL 9, Windows Server 2022, OpenSCAP Oct 2026

- Developed patch-window scheduling based on historical batch-job runtime percentiles (p95), with nonproduction-first sequencing, soak periods, and alerts for runtime growth above 50%.
- Automated dependency-ordered patching and reboots across four Terraform-provisioned hosts; performed 23 post-patch checks, OpenSCAP STIG scans, and S3 report archival.
- Resolved an SELinux-related post-patch nginx 502 error by correcting policy rather than disabling SELinux; pinned Java/Tomcat versions to validated releases. GitHub Repo

Multi-Region Disaster Recovery Active-Passive & Active-Active | AWS, Terraform, Python, Step Functions, FIS, GitHub Actions Oct 2026

- Architected a 118-resource, multi-region AWS disaster recovery environment using Terraform, Aurora Global Database, DynamoDB global tables, and Route 53, implementing automated active-passive failover orchestration with AWS Step Functions.

- Engineered automated chaos testing and RTO/RPO validation using AWS FIS and Python, achieving 129-second client-observed recovery with zero observed Aurora data loss, while identifying a 33-second DynamoDB replication gap and remediating failover automation defects.Github Repo

AWS Systems Manager Patch Automation & Security Hardening | AWS Systems Manager, EC2, CloudFormation, IAM, Linux, Windows Server Oct 2026

- Provisioned a four-instance Linux/Windows lab with SSM-only access, no inbound SSH/RDP, IMDSv2, custom patch baselines, and patch groups.
- Reduced missing patches from 164 to zero and improved reported compliance from 1/4 to 3/4 instances; configured maintenance windows and investigated OS-specific compliance discrepancies.GitHub Repo YouTube

Azure Disaster Recovery Drills Sep 2026

Azure Site Recovery, Azure SQL failover groups, Terraform, Azure CLI

- Tested Azure VM failover (2m34s against 15-minute RTO) and Azure SQL failover (12.9s against 60-second target); measured ~2.3-second database RPO.
- Modeled AWS hybrid connectivity using Site-to-Site VPN/IPsec and Transit Gateway; diagnosed routing, forwarding, and security-boundary failures and codified infrastructure in Terraform. (GitHub Repo)

Hybrid Cloud Network Architecture June 2026

Terraform, AWS Transit Gateway, Site-to-Site VPN

- Architected hybrid AWS networking using Transit Gateway, VPCs, Site-to-Site VPN/IPsec, routing, and security groups to simulate secure on-premises connectivity.
- Automated network infrastructure with Terraform and resolved cross-layer connectivity issues involving Linux IP forwarding, routing, and AWS security group configurations. (GitHub Repo)

PROFESSIONAL EXPERIENCE

EpixDevo

Cloud Engineer (Contract) *July 2026 – Present*

- Architected an AWS serverless application using API Gateway, Lambda, DynamoDB, and Cognito; secured frontend delivery with CloudFront Origin Access Control, ACM TLS, and private S3 origins.
- Designed a two-region hybrid network simulation with AWS Site-to-Site VPN, virtual private gateways, and routing; isolated connectivity failures to Linux IP forwarding and security-group scope.
- Evaluated HashiCorp Vault and managed secrets options; reduced cross-account deployment risk by explicitly scoping infrastructure-as-code execution to intended AWS accounts.

Onslow County Government | Jacksonville, NC

Cloud Support Engineer *Feb 2026 – July 2026*

Technology Support Specialist *Sept 2024 – Feb 2026*

- Diagnosed escalated Tier I/II infrastructure and cloud-service issues across Windows, Microsoft 365, Active Directory/Entra ID, networking, and authentication; documented root causes and remediation steps.
- Administered identity lifecycle requests, group membership, permissions, and access troubleshooting; partnered with infrastructure teams on cross-system incidents and knowledge-base documentation.

CarolinaEast Health System | New Bern, NC

IT Operations Specialist *Aug 2023 – Sept 2024*

- Resolved 90%+ of 40–50 weekly Tier I/II incidents supporting a 1,000+ user environment; troubleshot DNS, TCP/IP, file shares, endpoint systems, and Microsoft 365 access.
- Handled 30+ weekly account provisioning, permission, deactivation, and authentication requests through Active Directory, Entra ID, and Microsoft 365.

EDUCATION

Western Governors Univ. | Bachelor of Science in Cloud Computing | Salt Lake City, UT May 2025

James Sprunt Comm. Col. | Associate of Science in Information Technology | Kenansville, NC May 2023

Per Scholas | AWS re/Start Training | Jacksonville, NC Nov 2026

15-week, full-time AWS re/Start training: cloud computing, networking, databases, and AWS management/monitoring/governance. Focus: Linux, Python, EC2, Lambda, Auto Scaling, Shell Scripting, DynamoDB.

CERTIFICATIONS

Linux Essentials Certification	Feb 2024
AWS Certified Solutions Architect – Associate	July 2025
Microsoft Certified: Azure Administrator Associate	July 2025
Microsoft Certified: Azure Fundamentals	July 2024
AWS Certified Cloud Practitioner	Jan 2024
CompTIA Security+	Nov 2023
CompTIA A+	Aug 2024